

Figure 2: A set of cards from different families

comment. The numbers that we wish to sort are represented as the key-select. By looking at the algorithm, you can see that the parameters are passed to a procedure by value.

As the algorithm is quite simple, it is self-explanatory. Now try to expand the same algorithm to another problem shown in Figure 2. If you have understood the first one, this will be quite simple, except that you need to bring in some additional rules as there are cards from different families and two cards here have the same value (of hearts and diamonds).

Analysing an algorithm has come to mean predicting the resources that the algorithm requires so as to get the desired output. It considers aspects like memory allocation, communication bandwidth, computer hardware, etc. We also need to take into account parameters like computational time when it comes to the practical side. These parameters may, in turn, depend on *input size* and the *running time*.

Methodology: the divide-and-conquer approach

This is an effective methodology that we can adopt when we design algorithms. It involves:

- Divide: Divide the given sequence (with n elements) into two sub-sequences (of $n/2$ elements each)
- Conquer: Sort the two new sub-sequences recursively using the merge sort algorithm.
- Combine: Merge the two sorted sub-sequences to produce the desired result.

The merge mode is illustrated below:

MERGE(A, p, q, r)

MERGE-SORT (A, p, r)

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if p < r                > To check for base case
then q < (p + r)/2      > For dividing
    MERGE-SORT (A, p, q) > Conquering
    MERGE-SORT (A, q + 1, r) > Conquering
    MERGE (A, p, q, r)  > Combining
  
```

The above pseudo code may not enlighten novice programmers, who can look at the expanded code given below and then use the above example to assimilate the core idea:

Some fields that rely on algorithms

Writing algorithms is not just the headache of programmers alone. There are various other fields in which we need to rely on the algorithmic approach:

- The Human Genome Project has the goal of identifying all the 100,000 genes in the human DNA. It has to determine about three billion sequences of the chemical base pairs! This is virtually impossible unless we employ effective algorithms for pattern recognition and identification.
- Today we depend a lot on electronic commerce for the purchase of any commodity. A good system should have the ability to keep information (such as credit card numbers, passwords and bank statements) secure and encrypted. Algorithms are used for these cryptographic processes.
- Other branches like physical science (see the box on 'Simulation building in physical science) require the use of algorithms for solving complex problems.
- Even a manufacturing industry (or any other commercial setting) needs algorithms for allocating scarce resources.

	1 Sec	1 Min	1 Hr	1 Day	1	1 Year
$n!$						
n^2						
$\ln(n)$						
$\log(n)$						
$n \times \ln(n)$						
2^n						

Table 1

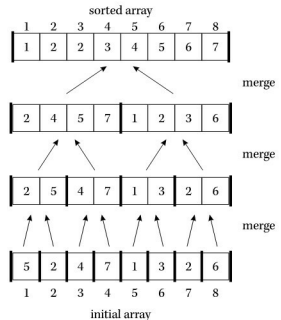


Figure 3: Sorting and arranging an array of values in numbers

MERGE (A, p, q, r)

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s1 ← q - p + 1
s2 ← r - q
  
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